TITLE: TO BE DECIDED

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Contents

1. Introduction

Abstract. In recent papers [1] [2], topological data analysis has proven to be a very effective method for analyzing spatiotemporal patterns in the spread of drug use and disease. We extend these results to the state of Virginia by using a combination of classic tools from statistics and recent tools from topological data analysis (or TDA) by developing a hybrid method for analyzing the evolution of drug overdose data for the state of Virginia. By using the Kepler mapper algorithm along with a multifaceted lens function, combining an isolation forest along with the l_2 norm and a K-Means clustering algorithm we analyze and identify the evolution of patterns and hotspots in the spatiotemporal distribution of the Virginia drug overdose data. Additional contextual variables, including county-level drug-related arrest rates, are incorporated to examine interactions between public health outcomes and enforcement patterns.

2. Data Sources

To account for variation in county population size across the Commonwealth of Virginia, we incorporated county-level population estimates for the year 2025 obtained from the Virginia Population Estimates Project. These estimates were sourced from the Weldon Cooper Center for Public Service and compiled in the dataset `virginia-counties-by-population-(2025).csv`. The dataset provides population estimates for all Virginia counties and independent cities.

Opioid overdose outcome data were obtained from the Virginia Department of Health (VDH) through multiple publicly available datasets. Annual county-level overdose mortality data for the years 2018–2023 were sourced from the dataset `VDHPUDOverdoseDeathsByYe`

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arAndGeographyLongLat.csv, which includes geographic identifiers and associated latitude and longitude coordinates. More recent overdose mortality data for 2024 and 2025 were obtained from vdh-pud-overdose-deaths-by-year-and-geography.csv, extending the temporal scope of the analysis.

Additional supporting datasets published by the Commonwealth of Virginia were used to contextualize overdose mortality trends. These include monthly counts of deaths by select causes (2014–2019), emergency department visits related to opioid overdose by health region and month, overdose deaths reported by year and quarter, arrest data with geographic identifiers, and indicators from the Virginia Drug Overdose Needs Assessment Tool. All datasets were obtained from the Virginia Department of Health and the Commonwealth of Virginia open data portal.

Arrest data were incorporated to provide a complementary measure of drug-related activity and enforcement patterns across Virginia. County-level arrest records were obtained from the Commonwealth of Virginia open data portal and compiled in the dataset GIS_Arrest_data.csv. This dataset includes geocoded arrest incidents with associated offense classifications, temporal information, and county-level geographic identifiers.

Drug-related arrests were extracted based on offense codes corresponding to controlled substance possession, distribution, and related violations. These data provide an indirect measure of drug market activity and law enforcement engagement, offering contextual information that may precede, coincide with, or lag behind observed overdose outcomes. To further characterize community-level vulnerability related to drug overdose, we incorporated locality-level scores from the Virginia Drug Overdose Needs Assessment Tool (ONAT), developed by the Virginia Department of Health. These data were obtained from the dataset vdh-pud-onat-scores-by-locality.csv, which provides composite overdose risk scores for Virginia counties and independent cities.

The ONAT scores synthesize multiple indicators associated with overdose risk, including measures related to overdose mortality, emergency department utilization, prescribing patterns, socioeconomic conditions, and access to treatment and harm reduction services. Scores are intended to support public health planning by identifying localities with elevated overdose-related needs relative to other jurisdictions within the Commonwealth.

These locality-level scores were used as contextual covariates in the analysis to complement observed overdose mortality outcomes. By incorporating ONAT scores, the analysis accounts for underlying structural and health-system factors that may influence overdose risk beyond observed death counts alone.

3. Data Preprocessing and Normalization

Raw opioid outcome measures were normalized using county-level population estimates to mitigate bias introduced by population size. For each county i and year t, let $O_{i,t}$ denote the observed opioid-related outcome and let $P_{i,t}$ denote the corresponding population estimate. Population-normalized rates were computed according to

$$R_{i,t} = \frac{O_{i,t}}{P_{i,t}} \times 10{,}000, \tag{1}$$

where $R_{i,t}$ represents the outcome rate per 10,000 residents. This normalization step ensures that downstream analyses reflect relative prevalence rather than absolute counts.

The overdose datasets span multiple temporal resolutions, including monthly, quarterly, and annual reporting intervals. To ensure consistency across datasets, monthly and quarterly measures were aggregated to annual summaries when integrated with county-level population data. Counties with missing observations for specific years were retained when possible to preserve spatial continuity.

All normalized variables were subsequently scaled prior to topological analysis to ensure comparability across dimensions and to prevent any single variable from dominating the distance metric.

3.1. **Arrest Data Processing.** Let $A_{i,t}$ denote the total number of drug-related arrests recorded in county i during year t. To ensure comparability across counties with varying population sizes, arrest counts were normalized using county-level population estimates. Population-adjusted arrest rates were computed as

$$A^*_{i,t} = \frac{A_{i,t}}{P_{i,t}} \times 10{,}000, \tag{2}$$

where $P_{i,t}$ denotes the estimated population of county i in year t. The resulting arrest rate $A^*_{i,t}$ represents the number of drug-related arrests per 10,000 residents.

Temporal aggregation was performed to align arrest data with the annual resolution of overdose mortality data. Arrest records spanning multiple months within a calendar year were aggregated to yearly totals prior to normalization. Counties with incomplete arrest reporting were retained where possible to preserve spatial coverage.

Incorporating arrest data alongside overdose mortality enables examination of the interaction between public health outcomes and law enforcement activity. While overdose deaths reflect acute health consequences of drug use, arrest rates capture behavioral and institutional responses associated with drug availability, enforcement intensity, and regional policy differences.

From a topological perspective, arrest rates introduce an additional dimension that may influence connectivity and clustering within the data. Regions with similar overdose mortality profiles may exhibit distinct arrest patterns, potentially revealing latent structural features when embedded in a higher-dimensional feature space.

In subsequent analyses, normalized arrest rates were incorporated either as auxiliary covariates or as components of lens functions within the Kepler Mapper framework. This allows for exploration of how enforcement intensity co-evolves with overdose patterns and whether persistent topological features are sensitive to variations in arrest activity.

4. Topological Data Analysis Framework

Each county-year observation was represented as a point in a multidimensional feature space consisting of population-normalized overdose mortality rates, selected public health indicators, and geographic coordinates. Latitude and longitude information enabled counties to be embedded directly in a two-dimensional spatial framework, allowing geographic proximity to influence the underlying metric structure.

Distance between observations was computed using the standardized feature vectors, and the resulting metric space served as input for topological data analysis. Population normalization plays a critical role in the topological representation, as persistent homological features are sensitive to scale and distance relationships. By using per-capita opioid rates, the resulting topological structures capture similarities in epidemic severity rather than artifacts driven by county population size.

Persistent homology was applied to this metric space to identify connected components, cycles, and higher-dimensional features that characterize structural patterns in the opioid epidemic across Virginia. Incorporating both spatial and temporal information enables identification of regional clustering, transitional zones, and persistent hotspots that may not be apparent through traditional statistical or regression-based approaches.

Data Availability

Population estimates were obtained from the Weldon Cooper Center for Public Service at the University of Virginia and are publicly available at https://www.coopercenter.org/virginia-population-estimates. Opioid overdose mortality and related public health datasets were obtained from the Virginia Department of Health and the Commonwealth of Virginia open data portal, including county-level overdose deaths by year and geography, emergency department visit data, arrest data, and indicators from the Virginia Drug Overdose Needs Assessment Tool.

References

[1] Nicholas Bermingham, David White, Nathan Willey *Spatiotemporal Spread of the Ohio Overdose Epidemic with Topological Data Analysis.*

[2] Yiran Chen, Ismar Volić *Topological data analysis model for the spread of the coronavirus.* PLoS ONE; 2021;16(8):e0255584. doi:10.1371/journal.pone.0255584.

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